

Cooperative Multi-Agent Jamming of Multiple Rogue Drones Using Reinforcement Learning

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Abstract—The wide adoption and use of unmanned aerial vehicles (UAVs) has created not only opportunities but also threats to the security of sensitive areas. Thus, effective and efficient counter-drone systems are required to protect these areas. This work tackles this issue by developing cooperative multi-agent jamming techniques using reinforcement learning (RL) to counter the operation of one or multiple rogue drones flying over a sensitive area. The aim of the proposed RL approach is to optimize the joint mobility and power control actions of the pursuer UAVs in order to maximize the received jamming power at the rogue drones aiming at disrupting communication links and sensing circuitry, while at the same time keeping the interference to surrounding pursuer agents below a predefined threshold. The effectiveness of the proposed approach in terms of scalability, learning speed, and agents' final joint performance is demonstrated through extensive simulation experiments for various agent and target configurations.

Index Terms—Multi-agent systems, jamming, reinforcement learning, q-learning, optimization.

I. INTRODUCTION

THE interest for drones has significantly increased over the recent years due to the technological advancements, miniaturization, and the wide range of applications they offer. The Federal Aviation Administration (FAA) predicts that approximately 1.84 million drones will be operating in the USA before the end of 2026 [1]. Additionally, the global drone market is expected to grow to 41.3 billion USD by the same year [2]. Despite their potential benefits, drones pose new risks and challenges to security and public safety, particularly when they fly over critical infrastructures and public spaces [3], [4].

A range of methods for countering rogue drones have been proposed over the recent years, focusing on detection and interception techniques. However, these methods alone are

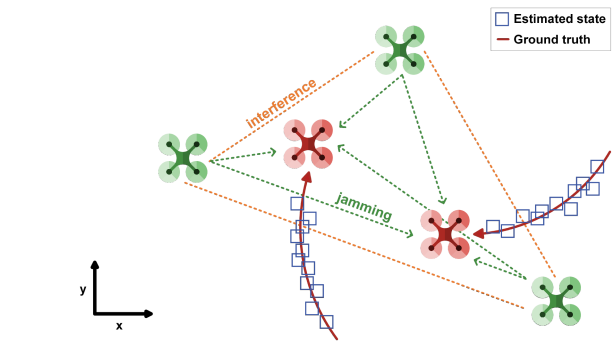


Fig. 1. Cooperative jamming multiple rogue drones.¹

insufficient, and further work is necessary to eliminate the potential safety threats posed by drones [5]. For instance, while radio-frequency (RF) jamming is the most employed defense measure in commercial and military anti-drone systems [6], its use for non-military applications is restricted by many national regulations due to the possibility of interfering with existing communication systems.

This work examines the deployment of a team of UAV agents for cooperatively jamming multiple rogue drones in the air with the purpose of disrupting their communication links and sensing receivers, while simultaneously maintaining minimal wireless interference to surrounding communication systems. A realistic scenario is considered where the rogue drones move with uncertain and noisy dynamics and the UAV agents are employed with a range-finding sensor that returns noisy target measurements. Additionally, the UAV agents exhibit a limited sensing range and are able to detect the existence of a rogue drone within their sensing range with probability less than one. Except for rogue drone measurements, the UAV agents additionally receive false-alarm measurements, since the range-finding sensor is assumed to be imperfect. Fig. 1 illustrates the problem tackled in this work.

The cooperative multi-agent problem of jamming multiple rogue drones is first formulated as a constrained optimization problem which nevertheless is intractable to solve due to its high complexity, especially for a large number of agents and targets. Consequently, the problem is formulated as a Markov Decision Process (MDP) and is being addressed using reinforcement learning.

¹Drone icons made by Freepik [https://www.flaticon.com/authors/freepik] from www.flaticon.com.

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The proposed learning framework, that extends our work in [7], is able to be executed in real-time and is scalable with respect to the number of rogue drones and UAVs. Compared to [7], this work significantly improves the coordination between the agents, in order to enhance the scalability, the learning speed and the overall system performance, by introducing a more informative reward function that deals with the credit assignment problem by accounting for the individual contribution of each UAV agent. Further, contrary to [7], this work considers both the cases of omnidirectional and directional antennas for jamming the rogue drones, as well as the case where an evader target is considered that when jammed it alters its mobility dynamics depending on the received jamming power. Specifically, the main contributions of this work are as follows:

- The problem of cooperative jamming of multiple rogue drones is formulated in challenging conditions, e.g., detection uncertainty, limited sensing range, false-alarms, and unknown rogue drone trajectories. The system model is also extended to consider both directional and omnidirectional antennas as well as non-evasive and evasive targets.
- The problem is formulated as an MDP, and an approach based on reinforcement learning, specifically Q-learning, is proposed. Such an approach allows a team of UAV agents to optimize their joint mobility and power control actions at each time-step in order to maximize the received power at the rogue drones, while keeping the interference power received at the UAVs below a critical threshold.
- A number of reward functions are discussed for either maximizing the jamming objective or minimizing the interference power and a joint novel reward function is designed that encourages the coordination between the learning agents to closely achieve the constraints and the objectives of the cooperative jamming problem.
- Extensive simulation experiments demonstrate the effectiveness of the proposed approach, for various agent and target configurations, in terms of improved scalability, learning speed, and agents' final joint performance.

The rest of the paper is organized as follows. Section II overviews the related work, while an introduction to RL is provided in Section III. Section IV presents the system model and defines the cooperative jamming problem. Section V formulates the proposed RL-based multi-agent control approach for cooperative jamming, while Section VI presents an extensive performance evaluation. Finally, Section VII concludes the paper and discusses possible future research directions.

II. RELATED WORK

Multiple works have examined the state-of-the-art counter-drone systems. The survey in [8] summarizes detection and identification technologies for drones, such as radar, RF, acoustic, optical, and thermal, and overviews neutralization technologies, including jamming, spoofing, hijacking, and capturing. The authors in [9] classify the counter-drone systems as either ground- or sky-based solutions, while the authors in [6], [10] survey numerous commercial anti-drone systems. Moreover, the work in [4] addresses the challenges of defending airports from rogue

drones and examines the benefits and drawbacks of the existing countermeasures. Further information on drone detection, tracking, and mitigation techniques is available in [11], [12], [13].

In addition, several works, e.g., [10], [14], [15], implement ground-based anti-drone systems for detecting and RF jamming a rogue drone. When a rogue drone is detected, a powerful jamming signal is transmitted towards it, causing it to disconnect from its controller and activate its failsafe mode. Regarding drone detection mechanisms, the approach in [14] uses a radar, while [15] uses RF analysis, and [10] fuses audio signals, computer vision, and RF signals. Further, more recent works in [16], [17] present a prototype passive radar system (deployed on the ground and onboard the UAV) which is based on signals of opportunity and software-defined radio, for detecting-and-tracking malicious drones operating over sensitive regions. Moreover, the authors in [18], [19] propose and develop aerial counter-drone systems where an autonomous drone detects, tracks, and jams a rogue drone. Additionally, in [19] the pursuer drone also achieves self positioning as it switches from the jamming to the self positioning module whenever its own localization is impacted due to the jamming signal. Finally, a recent work in [20] develops a cooperative counter-drone system that utilizes both ground- and sky-based technologies, while testing its applicability with field tests. The developed framework can deal with a single rogue drone at a time, but can be scaled to a large number of rogue drones over time by including more interceptor and pursuer UAVs (i.e., UAVs capable to intercept malicious drones and UAVs that monitor the airspace and can verify a possible target, respectively), where each interceptor-pursuer pair handles a single rogue drone at a time.

The use of cooperative teams of UAVs can enhance the effectiveness and efficiency of some neutralization technologies, or in the case of collision UAVs (i.e., dedicated UAVs that neutralize malicious drones by colliding with them), the team itself is the neutralizer. A recent review in [21] investigates the advantages of these systems, i.e., cooperative drone-based counter-drone systems, that include mobility, flexibility of deployment, scalability, and coverage expansion. The authors in [22] present a dynamic radar network, comprised of UAVs employed with radars, for detecting and tracking a rogue drone; nevertheless, no defense technique is considered. Further, the work in [23] proposes an approach for catching a rogue drone in the air utilizing a net carried by cooperative UAVs; however, the presence of a reliable target detection system is assumed. Moreover, the work in [24] proposes an approach where multiple networked UAVs are positioned in an enclosing formation around a rogue drone to restrict its movements and force it to move outside a critical area; the rogue drone, however, is assumed to be avoiding collisions with the rest of the UAVs and a highly accurate drone detection-and-tracking system is additionally assumed. Finally, the works in [25], [26] examine the problem of tracking-and-jamming rogue drones by multiple cooperative UAVs in 2D environments, and propose centralized and distributed decision and control algorithms for addressing this problem. In particular, in [26] a centralized approach is proposed where the team of UAVs jointly decides upon jamming power level and mobility control

actions so that the jamming power received at the rogue drone is maximized, whereas in [25] a distributed control approach is proposed for the same problem. In a similar vein, in [27] the authors develop a distributed estimation and control technique for tracking-and-jamming a malicious drone utilizing a group of pursuer agents that maneuver in 3D space while avoiding collisions with obstacles within that area, and while also adhering to jamming interference constraints. A prototype implementation of a counter-drone system utilizing a group of pursuing agents was demonstrated in [28], where an SDR-based system is implemented that allows for jamming the rogue drone and also self localizing (through a cooperative positioning framework based on SOPs as well as inertial and vision measurements).

In recent years, many UAV-related tasks have been solved with RL, including drone racing [29], UAV-aided cellular networks [30], UAV coverage path planning [31], and UAV state identification and trajectory prediction [32]. Moreover, in [33], a UAV learns a maneuver for pursuing an evader drone, while in [34] and [35] multiple pursuers learn to capture a single superior evader, with [35] assuming poor communication conditions. In [36], multiple UAVs learn to control their trajectories for tracking multiple ground targets in challenging environments. Finally, in [37], a ground drone defense system is proposed that is employed with various defense mechanisms and uses Q-learning to learn an optimal defense policy.

This work complements the state of the art as it provides RL-based cooperative jamming of multiple rogue drones that allows the pursuer UAVs to optimize their joint mobility and power control actions so as to maximize the jamming objective (i.e., jamming power received at the rogue drones) and at the same time minimize the interference power experienced by the pursuer drones (i.e., keeping it below a set threshold).

III. BACKGROUND

Reinforcement learning [38], a paradigm of machine learning, is concerned with an agent learning through interactions with its environment so as to maximize a numerical feedback known as the reward. A trial-and-error approach is utilized to collect this feedback where the agent tries out different actions and receives responses from the environment, i.e., rewards, relative to its actions. The goal of a reinforcement learning problem is to find an optimal policy, i.e., to map states to actions, so as to maximize the accumulated reward.

Additionally, a Markov Decision Process is a mathematical framework used for defining reinforcement learning problems. A MDP consists of a 5-tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, where $\mathcal{S}$ and $\mathcal{A}$ denote the set of all states and actions respectively; $\mathcal{P}(s, a, s')$ is the transition probability function that indicates the probability of making a transition from state s to s' by taking action a , i.e., $\mathcal{P}(s, a, s') = \Pr(s_{k+1} = s' | s_k = s, a_k = a)$; $\mathcal{R}(s, a)$ is the reward function that denotes the expected immediate reward received when taking action a from state s , i.e., $\mathcal{R}(s, a) = \mathbb{E}(r_k | s_k = s, a_k = a)$; and $\gamma \in [0, 1]$ is the discount factor which represents the importance of future rewards. A policy $\pi(s, a) = \Pr(a_k = a | s_k = s)$ is a mapping from states to actions and denotes the probability of taking action a from state s .

A popular RL algorithm is Q-learning, where an optimal policy is learned iteratively by estimating the values of state-action pairs, $Q(s, a)$, and storing them in a table, denoted as the Q-table. The Q-learning update is given by

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (1)$$

where α is the learning rate and r is the reward received for transitioning from state s to new state s' by taking action a .

Since many RL applications are subject to large or continuous state and action spaces, they often require function approximation. Tile coding [38] is a common linear function approximator that exhaustively partitions the state space into a set of tilings and tiles.

A cooperative multi-agent system refers to a system of multiple decision-making agents, placed in a common environment, which interact with each other towards achieving a common goal [39]. Typically, such a common objective is complex, and multi-agent RL (MARL) is a promising way to address the emerging complexity. However, various new challenges arise for RL in multi-agent systems [40], including (i) partial observability experienced by the agents, i.e., imperfect information about the system state s and action taken a ; (ii) scalability of the algorithms with respect to the number of learning agents; and (iii) difficulty in determining the contributions of a particular agent to a common task, i.e., the credit assignment problem.

Independent learners (ILs) and joint action learners (JALs) are two typical approaches to deal with MARL [41]. In the case of multiple ILs the assumption is that from each specific agent's perspective all other agents are part of the environment and, therefore, as they learn concurrently, the environment appears to be non-stationary from that agent's perspective. In particular, the action taken by an agent influences the reward of other agents, and the evolution of the state. To deal with the non-stationarity, JALs account for the actions taken by the other agents by learning in the joint action space which, however, scales exponentially with the number of agents, rendering JALs inappropriate in practice. Therefore, in this work the focus is on multiple ILs and not JALs, as scalability and minimal communication between agents are of utmost importance.

The direct application of ILs, however, also poses several challenges, and thus it is used in conjunction with other mechanisms. Specifically, communication tackles the problem of partial observability [42], function approximation, such as tile coding, addresses the scalability problem, and specialized reward functions (such as difference rewards) addresses the credit assignment problem.

Typically, two reward functions exist in MARL problems. A *local reward* L_j is distinct to each agent and usually encourages selfish behaviors as each agent seeks to maximize its own reward, potentially at the expense of the system's performance. The *global reward* G represents the system's performance and motivates the agents to act in the system's interest; however, in multi-agent environments, the global reward typically contains noise associated with other agents acting in the system. That is, an agent which has taken a good action may receive a penalty because one or more of the other agents

have taken a bad action; this is termed the multi-agent credit assignment problem. Therefore, the global reward allows a large amount of noise to be inserted into the signal offered by the reward function, and as the signal-to-noise ratio decreases, so does the agent's performance. This multi-agent credit assignment problem can be tackled with difference rewards, that is a form of reward shaping [43] as described below.

The *difference rewards* mechanism rewards an agent based on its contribution to the system's task by removing a large amount of the noise created by the actions of the other agents in the system. The difference reward D_j is a shaped reward signal that quantifies each agent's individual contribution to the system performance and exhibits some theoretical properties [43], one of which is the property of sensitivity. In particular, D_j is sensitive to the actions of agent j ; an agent taking good actions should receive a higher reward, while an agent taking poor actions should receive a lower reward. Since the difference rewards signal D_j only depends on the actions of agent j , D_j provides a cleaner signal by removing a large amount of the noise created by the actions of other agents active in the system. The difference reward D_j is given by [44]

$$D_j(s, a) = G(s, a) - G(s_{-j}, a_{-j}), \quad (2)$$

where $G(s, a)$ is the global reward and $G(s_{-j}, a_{-j})$ is the counterfactual term, i.e., the global reward for a theoretical system without the contribution of the j th agent.

IV. PROBLEM DEFINITION

A. Target Dynamics

A set of rogue drones (i.e., targets), denoted by $I = \{1, 2, \dots, M\}$, move in the environment with dynamics described by the following discrete-time dynamical model:

$$x_k^i = \zeta(x_{k-1}^i) + \Gamma \nu_k, \quad (3)$$

where $x_k^i = [x^i, \dot{x}^i, y^i, \dot{y}^i, \omega^i]^\top \in \mathbb{R}^5$ is the state vector of the i th target at time-step k containing the position (x^i, y^i) and velocity $(\dot{x}^i, \dot{y}^i)$ in 2D Cartesian coordinates, as well as turn rate ω^i . The assumption in this work is that the targets/agents are maneuvering within the 2D x-y plane (i.e., focusing only on 2D trajectories instead of 3D), as operating in a 3D world model increases significantly the problem space without providing any additional key insight on the solution that will be demonstrated later on. The extension to a 3D model is noted as a possible future work.

The process noise $\nu_k \sim \mathcal{N}(0, \Sigma_\nu)$ models the random disturbances in the evolution of the state and follows a normal distribution with zero mean and covariance matrix $\Sigma_\nu = \text{diag}[\sigma_x^2, \sigma_y^2, \sigma_\omega^2]$. The non-linear function $\zeta(x)$ and the Γ matrix are given by the coordinated turn model [45], [46]

$$\zeta(x) = \begin{bmatrix} x + \frac{\dot{x}}{\omega} \sin(\omega \Delta T) - \frac{\dot{y}}{\omega} (1 - \cos(\omega \Delta T)) \\ \dot{x} \cos(\omega \Delta T) - \dot{y} \sin(\omega \Delta T) \\ y + \frac{\dot{y}}{\omega} (1 - \cos(\omega \Delta T)) + \frac{\dot{x}}{\omega} \sin(\omega \Delta T) \\ \dot{x} \sin(\omega \Delta T) + \dot{y} \cos(\omega \Delta T) \\ \omega \end{bmatrix}, \quad (4)$$

$$\Gamma = \begin{bmatrix} \frac{\Delta T^2}{2} & 0 & 0 \\ \Delta T & 0 & 0 \\ 0 & \frac{\Delta T^2}{2} & 0 \\ 0 & \Delta T & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (5)$$

where ΔT is the sampling interval. The target dynamics are assumed to satisfy the Markov property as shown in (3).

It should be noted that the proposed framework does not depend on this specific target dynamics model, i.e., does not depend on the coordinated turn model (for example, as shown later in Section VI-C, a different model is utilized for the target dynamics that reacts to jamming). The only limitation for the model of target dynamics is for the function $\zeta(x)$ to satisfy the Markov property, so that the Bayes recursion can be used to extract the estimated target states. The indicated model, i.e., the coordinated turn model, is chosen to illustrate a usual trajectory that could be followed by rogue drones especially when doing reconnaissance missions to gather information around certain areas. It also illustrates the general scenarios where non-linear paths are followed (instead of simplistic straight lines). This demonstrates the applicability of the proposed model for realistic scenarios with non-linear trajectories. An alternative scenario where the indicated model is used in the literature includes the work in [47], where a pursuer attempts to capture an evader while being constrained to circular motion, and the work in [48], where a target that follows a circular trajectory is being tracked.

In terms of how the rogue drones are identified, it is assumed that the cooperating agents have preset and agreed identification information broadcasted to uniquely identify each agent. The reader should note that this issue has already been addressed by the European Union Aviation Safety Agency (EASA), that plans the utilization of a remote identification system in order to identify all cooperating (friendly) agents and consequently also being able to identify non-cooperative (rogue) agents [49].

B. Agent Dynamics

A team of controllable UAV agents, represented by $J = \{1, 2, \dots, N\}$, cooperate to counter the operation of the rogue drones. Each UAV agent $j \in J$ follows the discrete-time dynamics given by

$$\bar{s}_k^j = \bar{s}_{k-1}^j + a_{l_1, l_2} \bar{s}_{k-1}^j + \begin{bmatrix} \Delta_R[l_1] \cos(l_2 \Delta_\theta) \\ \Delta_R[l_1] \sin(l_2 \Delta_\theta) \end{bmatrix}, \quad \begin{matrix} l_1 = 1, \dots, |\Delta_R| \\ l_2 = 0, \dots, N_\theta \end{matrix}, \quad (6)$$

where $\bar{s}_{k-1}^j = [\bar{s}_x^j, \bar{s}_y^j]^\top \in \mathbb{R}^2$ is the state vector of agent j (i.e., the position in 2D Cartesian coordinates) at time-step $k-1$, Δ_R is the vector of possible radial step sizes (with $\Delta_R[l_1]$ returning the value at index l_1), $\Delta_\theta = 2\pi/N_\theta$, and the parameters $(|\Delta_R|, N_\theta)$ determine the number of possible mobility control actions. The mobility control action a_{l_1, l_2} represents the direction and step size of the movement along the x and y axes. At each time-step k , each UAV agent chooses one mobility control action from the discrete set $\mathbb{U} = \{a_{l_1, l_2} \mid l_1 = 1, \dots, |\Delta_R|, l_2 = 0, \dots, N_\theta\}$, as computed by (6), while

agent j 's set of all admissible mobility controls is denoted by $\mathbb{U}_k^j = \left\{ \bar{s}_k^{j,1}, \bar{s}_k^{j,2}, \dots, \bar{s}_k^{j,|\mathbb{U}_k^j|} \right\}$.

C. Agent Sensing Model

The UAV agents detect nearby targets with a limited sensing range, modeled by function $p_D(x_k, \bar{s}_k)$ which gives the probability of whether a target at state x_k and position Hx_k (where H is a matrix that extracts the (x, y) coordinates from a target's state vector) can be detected by an agent at state $\bar{s}_k$ at time-step k . This probability is modeled according to

$$p_D(x_k, \bar{s}_k) = \begin{cases} p_D^{\max} & \text{if } d_k < R_0 \\ \max\{0, p_D^{\max} - \eta(d_k - R_0)\} & \text{if } d_k \geq R_0 \end{cases}, \quad (7)$$

with $d_k = \|Hx_k - \bar{s}_k\|_2$ denoting the Euclidean distance between the agent and the target at time-step k , $p_D^{\max}$ denoting the detection probability for a target that is positioned within distance R_0 from the agent, and parameter η indicating the reduced effectiveness of the agent to detect distant targets. The reader should note that any generic ranging sensor has such a behavior, i.e., ranging sensors can be described by the probability model utilized in this work. For instance, a LiDAR sensor has a certain field of view (FOV) within which for a given integration time the number of samples collected reduce moving away from the center of the FOV [50]. At the edge of the FOV the detection drops rapidly to zero which is what our model indicates. The same behavior is experienced by radar systems as well.

Each agent is equipped with a 2D range-finding sensor which receives noisy heading and range measurements $y_k \in \mathbb{R}^2$ from the detected target with state x_k according to the measurement model given by

$$y_k = h(x_k, \bar{s}_k) + w_k, \quad (8)$$

where $h(x_k, \bar{s}_k) = \left[\arctan\left(\frac{\bar{s}_y - y}{\bar{s}_x - x}\right), \|\bar{s}_k - Hx_k\|_2 \right]^\top$ and the measurement noise $w_k \sim \mathcal{N}(0, \Sigma_w)$ follows a zero mean normal distribution with covariance matrix $\Sigma_w = \text{diag}[\sigma_\phi^2, \sigma_\zeta^2]$. The standard deviations $(\sigma_\phi, \sigma_\zeta)$ are range dependent, and given by $\sigma_\zeta = \zeta_0 + \beta_\zeta \|\bar{s}_k - Hx_k\|_2^2$ and $\sigma_\phi = \phi_0 + \beta_\phi \|\bar{s}_k - Hx_k\|_2$, with β_ζ and β_ϕ denoting scaling factors (in units of m^{-1} and rad/m , respectively).

Moreover, this range-finding sensor consists of M sensing slots for detecting each one of the M targets. As sensors are imperfect, the agents additionally receive false-alarm measurements $c_k^1, \dots, c_k^n$ with an average Poisson rate λ at each sensing slot of time-step k . These measurements follow the uniform distribution over the measurement space and are distributed based on the density function $p_c(c)$ (Section V-B explains in more detail how the target state is estimated, accounting also for the case of receiving false-alarm measurements).

Overall, each agent $j \in J$, at each time-step k and sensing slot i , receives the measurement set Υ_k^{ji} as follows:

$$\Upsilon_k^{ji} = \bigcup \left\{ a \in \{\emptyset, y_k^{ji}\}, b \subseteq \{c_k^1, \dots, c_k^n\} \right\}, \forall i \in [1, \dots, M], \quad (9)$$

where y_k^{ji} denotes the measurement received by agent j at sensing slot i , i.e., received from target i .

It should be noted that the range-finding sensors are assumed to be imperfect, with the factors representing this imperfection being noisy measurements, false-alarm measurements (i.e., measurements that do not correspond to rogue drones), and absence of rogue drone measurements. Clearly, the accuracy and reliability of the range finding sensors can be enhanced with compensatory measures implemented in conjunction with sensor technologies. However, this is out of scope of this work and is not considered here (noted as future research).

D. Agent Power Transmission Model

Each agent is employed with omnidirectional antennas for jamming the rogue drones. At each time-step k , agent j 's transmitted power P_k^j varies over a discrete set of power levels $\mathbb{L}$. The received power at a target with state x_k from an agent with state $\bar{s}_k$ and transmit power P_k is subject to the free-space path-loss (FSPL) model (in dB) [12], [14]

$$FSPL(x_k, \bar{s}_k) = 20\log_{10}(\|\bar{s}_k - Hx_k\|_2) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right), \quad (10)$$

where f denotes the carrier frequency (in GHz) and c denotes the speed of light. The received power at the target is then computed as

$$P_{R,k} = P_k / \left(10^{FSPL(x_k, \bar{s}_k)/10}\right) = P_k \Lambda(x_k, \bar{s}_k). \quad (11)$$

The reader should note that the frequency where jamming signals are injected in the attack model is based on typical radio bands that are used for controlling commercial drones in Europe. These are the 2.4 GHz (2.400-2.483 GHz) and 5 GHz (5.725-5.850 GHz) bands, with the 5 GHz band mostly used for video transmission because of its higher bandwidth (see related work in [14]). Thus, in this work the 2.4 GHz band is used in the attack model as it is the most common frequency used by commercial drones.

1) Directional Antennas: As a next step, directional antennas are considered. The antenna profile is described by a quarter of a circle given by: $[x = u \sin \theta_a, y = u \cos \theta_a]$ for $u \in (0, r]$, $\theta_a \in \{(0, \pi/2], (\pi/2, \pi], (\pi, 3\pi/2], (3\pi/2, 2\pi]\}$, where r is the circular radius that characterizes the effective antenna range and θ_a is the angle that determines the antenna's quarter $\mathcal{L}$. It should be noted that due to the directional antennas, (11) is valid only when x_k is positioned inside quarter $\mathcal{L}$ of the antenna of the UAV agent. When the rogue drone x_k resides outside $\mathcal{L}$, it receives no power from the UAV agent.

In this work, the aim is to demonstrate the potential of antenna diversity with a simple sectoring technique that has been used extensively in wireless telecommunication systems; especially cellular. Evidently, introducing more complex beamforming techniques using antenna diversity could potentially improve the system performance; this is left as future work.

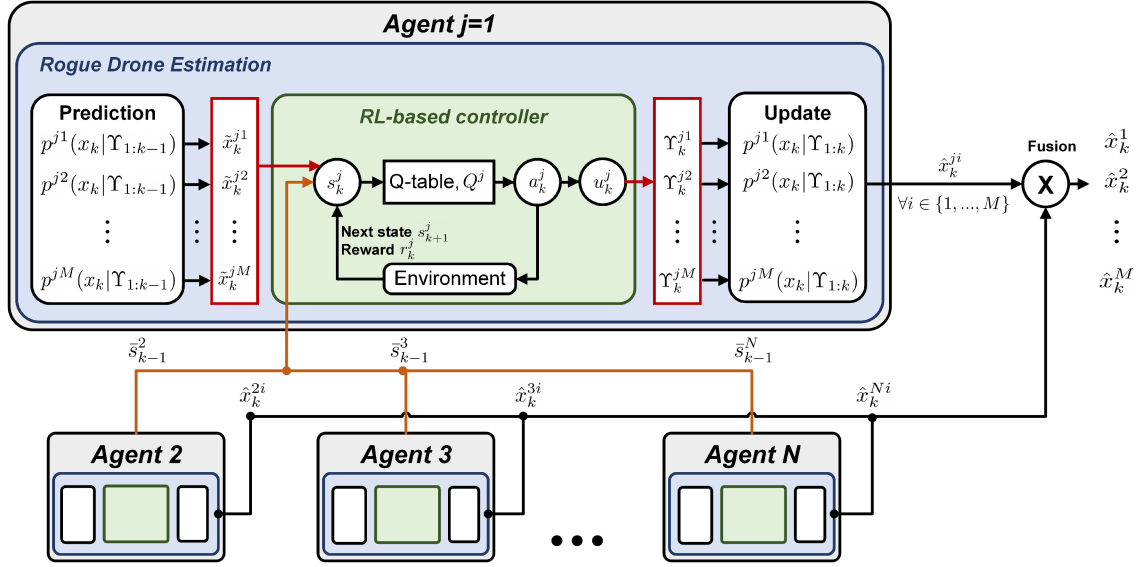


Fig. 2. System architecture.

E. Cooperative Jamming Objective

The problem of cooperative jamming multiple targets tackled in this work can be formulated as

$$(P1) \quad \max_{P_k^j, u_k^j} \min_{i \in I} \sum_{j=1}^N P_k^j \Lambda(\tilde{x}_k^{ji}, u_k^j) \quad (12)$$

$$\text{s.t.} \quad \sum_{j' \in J, j' \neq j} P_k^{j'} \Lambda(u_k^j, u_k^{j'}) \leq \Delta, \forall j \in J \quad (13)$$

$$P_k^j \in \mathbb{L}, u_k^j \in \mathbb{U}_k^j, \forall j \in J, \quad (14)$$

where the objective is to find the optimal mobility u_k^j and power P_k^j control inputs for each agent j , subject to the constraints in (13) and (14), that maximize the target-received jamming power, as indicated by the objective function in (12). More specifically, similar levels of jamming power received at the targets is imposed, by maximizing the minimum target-received power. Further, since the actual target state x_k^i is unavailable to the agents, each agent j maintains a prediction of the state of each target i , denoted by $\tilde{x}_k^{ji}$. The constraint in (13) ensures that no agent receives levels of interference power that are above a critical threshold, Δ , while the constraint in (14) restricts the agent control actions to the desired operation defined in Sections IV-B and IV-D.

The reader should note that, irrespective of the communication system utilized by the agents and the rogue drones, interference between cooperative agents should be reduced to maintain good signal quality. Moreover, and for each specific communication system (e.g., cellular, WiFi, RF), opting for high SNR between cooperating agents and low SNR for non-cooperating ones is preferable (imposed by the penalization factor in the objective of problem (P1)). Of course, other dedicated bands could be assumed by specific drones (i.e., the cooperative

agents and the rogue drones could potentially be using different frequencies, thus removing the need for Constraint (13); nevertheless, the assumption in this work is that commercial off-the-shelf (COTS) drones are used (also primarily used by law enforcement agencies) as well as open-source flight controllers, with these systems mainly using RF for telemetry, WiFi for data streams (e.g., camera feeds), and standard GNSS systems for localization (e.g., GPS, GLONASS, GALILEO). Thus, as potentially the cooperative and rogue agents, using COTS drones, could be using the same frequency, Constraint (13) is utilized in the formulation.

V. RL-BASED COOPERATIVE JAMMING

Fig. 2 illustrates an overview of the RL-based cooperative jamming system. In summary, this approach assigns a learning agent to each UAV that learns to coordinate with the other UAVs in the environment; however this is done without having the UAVs explicitly exchanging the learned knowledge amongst them. Specifically, each agent applies its own Q-learning algorithm, acquiring experience from acting in the environment, as shown in the green-shaded part of the figure. The agents only communicate their position of the previous time-step. Additionally, each agent uses stochastic filtering to estimate the state of the rogue drone recursively over time, as depicted in the blue-shaded part of the figure. Finally, the agents exchange their local estimation results in order to fuse them and produce the final target state estimates.

A. Proposed RL-Based Controller

In this section, the cooperative jamming problem is reformulated as an MDP (as discussed in Section III), which is then addressed using reinforcement learning.

State: The state vector of the j th agent at time-step k depends on the positions of the agents and the predicted position of each target i (which is extracted from the predicted target state $\hat{x}_k^i$), and is represented by $s_k^j = [m^{j1}, \dots, m^{j4}, p^{j1}, \dots, p^{j4}, n^{j1}, \dots, n^{j4}, q^{j1}, \dots, q^{j4}]_k$. Agent j 's state vector consists of four components:

- m^{jl} : number of targets in the l th area of agent j
- p^{jl} : average Euclidean distance between agent j and the targets in the l th area of agent j
- n^{jl} : number of agents in the l th area of agent j
- q^{jl} : average Euclidean distance between agent j and the remaining agents in the l th area of agent j

where $l \in \{1, 2, \dots, 4\}$. Essentially, at each time-step k each agent j splits the 2D space in four areas by assuming Cartesian axes centered at its current position coordinates and computes relative distances from the agents and targets. Notably, the state vector's size is constant and independent of the number of agents and targets and thus allows the framework to scale well with respect to the number of learning agents and targets, and the state vector is also independent of the environment's area size.

Action: At each time-step k , each agent j chooses an action a_k^j from the discrete action space $\mathcal{A} = \mathbb{U} \times \mathbb{L}$, that consists of a mobility and a power control action as defined in Sections IV-B and IV-D, respectively. Illegal actions are not allowed, e.g., an action that positions an agent outside of the environment's area.

Reward: At each time-step k , each agent j receives a scalar reward r_k^j from the environment, which is designed to closely approximate the objective and constraints of the optimization problem presented in Section IV-E.

The problem at hand consists of two conflicting objectives, i.e., maximizing jamming and minimizing interference. First, various rewards are investigated for maximizing the jamming objective:

(1) Local jamming reward for agent j , $L_{1,k}^j$, defined as the total jamming power received by the targets from agent j

$$L_{1,k}^j = \sum_{i=1}^M P_k^j \Lambda(x_k^i, u_k^j). \quad (15)$$

(2) Global jamming reward, $G_{1,k}$, defined as the summation of $L_{1,k}^j$ over all agents

$$G_{1,k} = \sum_{j=1}^N L_{1,k}^j = \sum_{j=1}^N \sum_{i=1}^M P_k^j \Lambda(x_k^i, u_k^j). \quad (16)$$

(3) Difference reward, $D_{1,k}^j$, for agent j for the jamming objective $G_{1,k}$, that can be computed by applying (2)

$$D_{1,k}^j = G_{1,k} - G_{1,k}^{(-j)}, \quad (17)$$

where the counterfactual term for jamming, $G_{1,k}^{(-j)}$, is calculated by assuming that the agent does not choose new mobility and power control actions for time-step k , as given by

$$G_{1,k}^{(-j)} = \sum_{\substack{j'=1 \\ j' \neq j}}^N \sum_{i=1}^M P_k^{j'} \Lambda(x_k^i, u_k^{j'}) + \sum_{i=1}^M P_{k-1}^j \Lambda(x_k^i, u_{k-1}^j). \quad (18)$$

(4) Global jamming variation reward, $G_{2,k}$, which ensures that each target receives adequate jamming power, i.e., the jamming power is uniformly distributed among the targets, and is defined as the minimum target received power

$$G_{2,k} = \min_{i \in I} \sum_{j=1}^N P_k^j \Lambda(x_k^i, u_k^j). \quad (19)$$

(5) Next, the objective of minimizing the interference power at the agents is considered. The global interference reward, $G_{3,k}$, is utilized as a penalty if the interference constraints in (13) are violated, and is defined as the maximum power received amongst all agents

$$G_{3,k} = \max_{j \in J} \sum_{\substack{j'=1 \\ j' \neq j}}^N P_k^{j'} \Lambda(u_k^j, u_k^{j'}). \quad (20)$$

Overall, each agent j receives the reward r_k^j as follows:

$$r_k^j = \begin{cases} R, & \text{if (13) is satisfied} \\ -G_{3,k}, & \text{otherwise.} \end{cases} \quad (21)$$

where if at least one agent receives interference power above threshold Δ , the agents are penalized with the global interference reward, $G_{3,k}$. Otherwise, the agents' reward comprises of R , that is either one of the jamming rewards $L_{1,k}^j$, $G_{1,k}$, $D_{1,k}^j$, and $G_{2,k}$. It should be noted that all rewards are normalized in $[0,1]$. The trade-off between the aforementioned reward functions is demonstrated below (Section VI).

RL Training: A summary of the training algorithm is provided in Algorithm 1. Each agent is an independent learner, having its own Q-table which is initialized with zero values (line 1). In each episode, lasting K consecutive time-steps, the environment is initialized randomly, encouraging the agents to learn various strategies via trial-and-error (line 3). Each agent selects an action according to an ϵ -greedy policy, i.e., selects an action that corresponds to the maximum Q-value in the current state with probability $1-\epsilon$ or to a random action with probability ϵ (line 6). The exploration parameter ϵ and learning rate α decay over time in order to encourage learning and exploration at initial episodes and exploitation afterwards (line 12). Each agent executes its action, receives the reward, and observes the new state before updating the Q-value of the current state-action pair (lines 7-9).

B. Rogue Drone State Estimation

The UAV agents need to estimate the target states recursively over time given noisy and false-alarm measurements. The objective of stochastic filtering [51] is to compute the conditional posterior distribution $p(x_k | y_{1:k})$ of a target state x_k at time-step k given measurements $y_{1:k} = y_1, y_2, \dots, y_k$ until time-step k . The prediction step and update step are given by the Bayes recursion

$$p(x_k | y_{1:k-1}) = \int p(x_k | x_{k-1}) p(x_{k-1} | y_{1:k-1}) dx_{k-1} \quad (22)$$

$$p(x_k | y_{1:k}) = \frac{p(y_k | x_k) p(x_k | y_{1:k-1})}{\int p(y_k | x_k) p(x_k | y_{1:k-1}) dx_k}, \quad (23)$$

Algorithm 1: Q-Learning Training for Multi-UAV Jamming of Rogue Drones.

```

1: Initialize Q-tables for each agent with zero values.
2: for episode  $n = 1, 2, \dots, N_e$  do
3:   Initialize environment.
4:   for timestep  $k = 1, 2, \dots, K$  do
5:     for agent  $j = 1, 2, \dots, N$  do
6:       Select a greedy action
          $a_k^j = \operatorname{argmax}_{a_k \in \mathcal{A}} Q^j(s_k^j, a_k)$  with
         probability  $1 - \epsilon$  or a random action with
         probability  $\epsilon$ .
7:       Take action  $a_k^j$ .
8:       Observe reward  $r_k^j$  and next state  $s_{k+1}^j$ .
9:       Update  $Q^j(s_k^j, a_k^j)$  using (1).
10:    end for
11:  end for
12:  Decay  $\alpha$  and  $\epsilon$ .
13: end for

```

where $p(x_k|x_{k-1})$ is the transitional density derived from (3), and $p(y_k|x_k)$ is the likelihood function.

Let us assume that each UAV agent j keeps a distinct density $p_k^{ji}(\cdot)$ for each target i , meaning that the number of targets is fixed and known, and the measurements' assignments to these targets are also known. At each time-step k , the agents receive a measurement set, so y_k becomes Υ_k^{ji} (modeled by a Random Finite Set (RFS) [52]), as discussed in Section IV-C. At the prediction stage, each agent j computes the predicted target state for each target i , using prediction density $p(x_k|y_{1:k-1})$ [obtained by (22)], given by

$$\tilde{x}_k^{ji} = \int x_k p^{ji}(x_k|\Upsilon_{1:k-1}) dx_k. \quad (24)$$

The posterior distribution $p(x_k|y_{1:k})$ depends on the measurement likelihood function $p(y_k|x_k)$ which in this work becomes $p(\Upsilon_k^{ji}|x_k^i, \bar{s}_k^j)$. As the UAV agents receive both noisy target and false-alarm measurements (discussed in Section IV-C), the measurement likelihood function of the j th agent and i th target is as follows:

$$p(\Upsilon_k^{ji}|x_k^i, \bar{s}_k^j) = \left[1 - p_D(x_k^i, \bar{s}_k^j)\right] e^{-\lambda} \prod_{\ell \in \Upsilon_k^{ji}} \lambda p_c(\ell) + e^{-\lambda} p_D(x_k^i, \bar{s}_k^j) \sum_{\ell \in \Upsilon_k^{ji}} p(\ell|x_k^i, \bar{s}_k^j) \prod_{\substack{\varepsilon \in \Upsilon_k^{ji} \\ \varepsilon \neq \ell}} \lambda p_c(\varepsilon), \quad (25)$$

where $p(y_k^{ji}|x_k^i, \bar{s}_k^j) = \mathcal{N}(y_k^{ji}; h(x_k^i, \bar{s}_k^j), \Sigma_\omega)$ is target i 's and agent j 's measurement likelihood function due to (8). To derive (25), the fact that $\Upsilon_k^{ji} = \{c_k^1, \dots, c_k^{|\Upsilon_k^{ji}|}\}$ is first used if the agent receives only false-alarm measurements (i.e., it does not receive any measurements from the target) and then $p(\Upsilon_k^{ji}|x_k^i, \bar{s}_k^j)$, is

computed as

$$p(\Upsilon_k^{ji}|x_k^i, \bar{s}_k^j) = |\Upsilon_k^{ji}|! \left[1 - p_D(x_k^i, \bar{s}_k^j)\right] \operatorname{Pois}(|\Upsilon_k^{ji}|) \prod_{\ell \in \Upsilon_k^{ji}} p_c(\ell), \quad (26)$$

where $\operatorname{Pois}(|\Upsilon_k^{ji}|) = \frac{\lambda^{|\Upsilon_k^{ji}|} e^{-\lambda}}{|\Upsilon_k^{ji}|!}$ is the Poisson distribution with rate parameter λ which gives the probability of obtaining exactly $|\Upsilon_k^{ji}|$ false-alarm measurements at time-step k and $\prod_{\ell \in \Upsilon_k^{ji}} p_c(\ell)$ is the joint density of the false-alarms, assuming they are independent. By adjusting this expression to consider all possible permutations of the measurement sequence and the fact that the target has not been detected, the first term of (25) is derived. The second term of (25) accounts for the case where agent j has detected target i , meaning that apart from false-alarm measurements, agent j has also received measurement y_k^{ji} .

At the update stage, i.e., after receiving measurement sets Υ_k^{ji} for time-step k , the posterior distribution is computed by (23) and the estimated target state $\hat{x}_k^{ji}$, is derived as

$$\hat{x}_k^{ji} = \int x_k p^{ji}(x_k|\Upsilon_{1:k}) dx_k. \quad (27)$$

Finally, the agents exchange their local estimation results in order to fuse them using covariance intersection [53] and obtain the final state estimate, $\hat{x}_k^i$, for each target i . More details on the derivation of the RFS measurement likelihood function and the estimated target state can be found in our previous work in [25].

C. Computational Aspects

The optimization problem (P1) is computationally expensive to solve using standard off-the-shelf solvers, since it exhibits exponential complexity with respect to the number of agents, i.e., $O((|\mathcal{U}||\mathcal{L}|)^N)$. This is due to the fact that the target received power of all possible combinations of actions needs to be calculated in order to determine the action combination that corresponds to the maximum target received power. On the contrary, the proposed RL-based controller provides a computationally cheap approach to address the same problem, as it is linear with respect to the number of the actions, i.e., $O(|\mathcal{U}||\mathcal{L}|)$. In the RL-based approach, the training phase typically consists of extensive simulations that require significant memory space and computational complexity (depending on the size of the state and action space), but this process is performed offline. Further, the prediction/inference stage only requires a lookup index of the Q-table which is learned during training, and in the proposed approach each agent has its own Q-table. Information exchanged between the agents is only limited and it is, therefore, not anticipated to affect a potential real-time deployment.

Apart from the performance trade-offs (discussed in Section VI), reward functions also differ in computational cost. More specifically, the difference reward $D_{1,k}^j$ assumes knowledge of $G_{1,k}$ (however it essentially reduces to knowledge of local information, i.e., it only depends on P_k^j , P_{k-1}^j , u_k^j , and u_{k-1}^j) and reward functions $G_{1,k}$ and $G_{2,k}$ assume communication between the agents to obtain P_k^j and u_k^j , $\forall j$. Reward function

TABLE I
SIMULATION PARAMETERS

Parameter	Value	Parameter	Value
ΔT	1 sec	$p_D^{\max}$	0.99
σ_x	2 m/s ²	η	0.003
σ_y	2 m/s ²	R_0	10 m
σ_ω	$\pi/180$ rad/s	Δ_R	[0, 1, 3] m
ϕ_0	$\pi/180$ rad	N_θ	4
β_ϕ	10^{-5} rad/m	$\mathbb{L}$	{off, -3, 0} dBm
ζ_0	1.5 m	f	2 GHz
β_ζ	3×10^{-4} m ⁻¹	Δ	-65 dBm
λ	10		

$L_{1,k}^j$ is the easiest to train, since it requires only local information but overall performs poorly.

VI. PERFORMANCE EVALUATION

A. Experimental Setup

The evaluation of the proposed approach is conducted using the following experimental setup.

Simulations: The UAV agents and the rogue drones operate in an area of size 100 m by 100 m, where for clarity no process noise is assumed in the actual target trajectory. The targets are randomly initialized inside the area, with x, y ranging from 0 to 100 m, and $\dot{x}, \dot{y}$, and ω taking values in the range of 0.20 to 0.40 m/s and 0.03 to 0.04 rd/s, respectively. The agents are randomly initialized to be within a 5 m radius around a target. A complete list of simulation parameters is presented in Table I.

RL: The agents learn for $N_e = 1, 6 \times 10^5$ episodes, where each episode is comprised of $K = 100$ time-steps. The Q-learning parameters are set as follows: the learning rate and exploration parameters decay linearly with each episode iteration, n , as given by $\alpha = 0.2 - 0.12 \frac{n}{N_e}$ and $\epsilon = 0.3 - \frac{n}{3.75N_e}$, respectively, while the discount factor is set to $\gamma = 0.9$. For representing the state space, each agent has its own tile coding function approximator comprising 2^6 grid tilings. The tile width in m^{jl} and n^{jl} dimensions is set to 1, while in p^{jl} and q^{jl} dimensions is set to 50 m. The values for the Q-learning and tile coding parameters are selected after performing parameter tuning.

Particle Filtering: For the implementation of the Bayes recursion in (22) and (23) particle filtering techniques are utilized [54].

Evaluation: The experiments are repeated over 50 independent runs. All training curve plots show average values across the 50 independent runs with a median filter of order 105 applied.

B. Performance Results

Initially, the proposed approach is evaluated by investigating the learning performance of the reward functions discussed in Section V-A. More specifically, the reward function in (21) is considered with R being either one of the following jamming rewards: the local jamming reward $L_{1,k}^j$, the global jamming reward $G_{1,k}$, the difference reward $D_{1,k}^j$, and the global jamming variation reward $G_{2,k}$. Fig. 3 illustrates the average target and agent received power per time-step

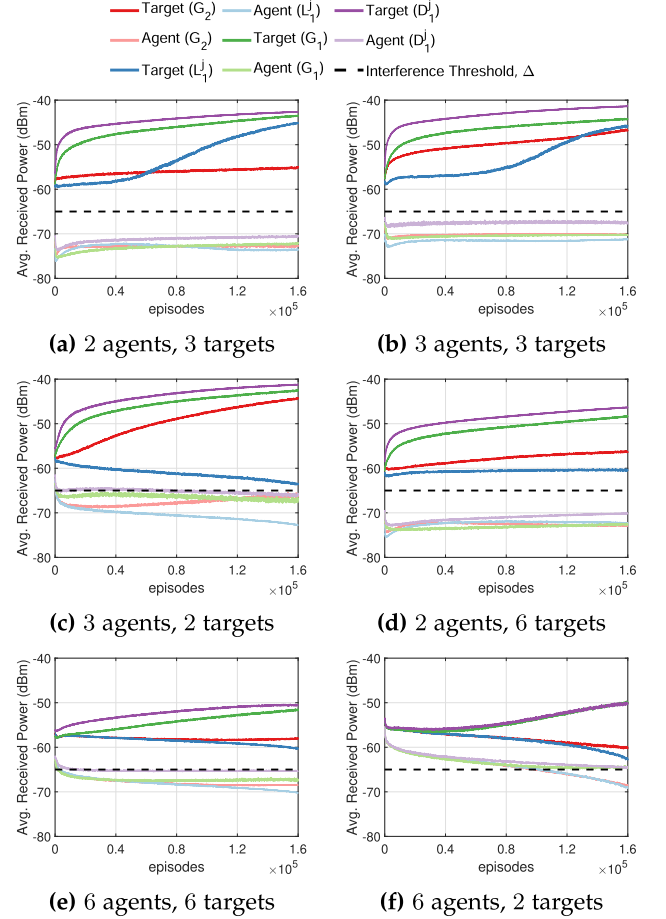


Fig. 3. Training curves for different number of agents and targets.

when the proposed system is trained for different number of agents and targets. For these experiments, it should be noted that target state estimation is not performed by the agents (i.e., the ground truth target states are assumed to be known) for ease of computation, especially for the larger scenarios with 6 agents and/or targets. As observed, the agent received power is kept below the interference threshold after a few initial episodes for all configurations. In terms of the target received power, as expected, $L_{1,k}^j$ performs poorly for most of the configurations, since the local reward encourages each agent to greedily maximize the power received at the drone that it is targeting, without considering the utility of the entire system. Reward function $D_{1,k}^j$ offers the best overall performance, while the global reward $G_{1,k}$ performs well but at a slower rate than $D_{1,k}^j$. Moreover, as demonstrated, $G_{2,k}$ performs well for a small number of targets and agents, but as their numbers increase the learning speed is very slow. This highlights the fact that $G_{2,k}$ alone (used in our previous work in [7]) is not informative enough and thus justifies the need for a more sophisticated reward function that deals with the credit assignment problem. Therefore, the experiments that follow utilize the difference reward $D_{1,k}^j$ as the reward function R .

The next experiment considers 3 agents cooperatively jamming 2 targets, while estimating the target states. It should be

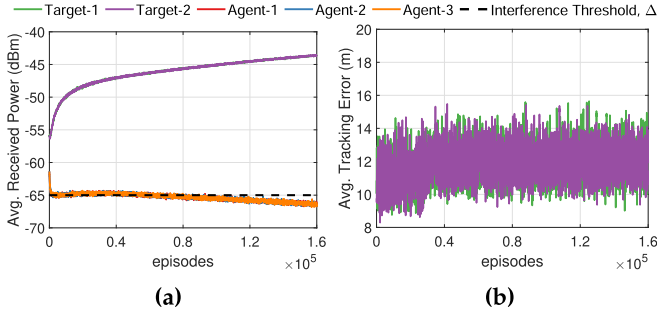


Fig. 4. Training curves for 3 agents and 2 targets.

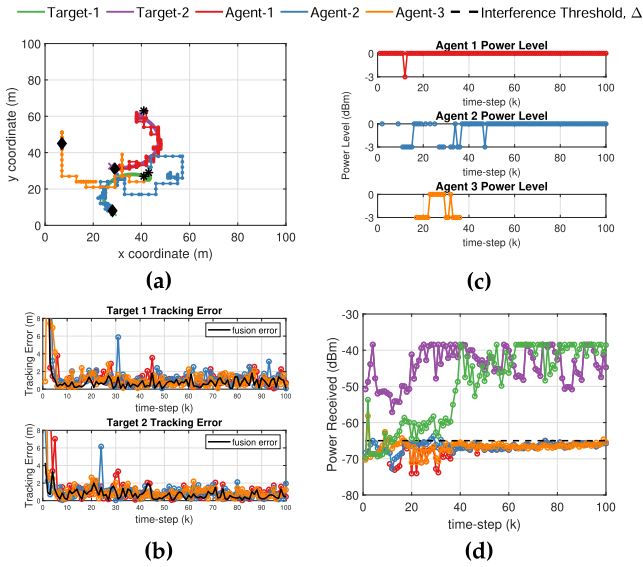


Fig. 5. A simulated scenario using 3 agents and 2 targets.

noted that in this case, the reward function $D_{1,k}^j$ utilizes the predicted target state, $\tilde{x}_k^{ji}$, $\forall i$, instead of the ground truth target state, x_k^i , $\forall i$. Fig. 4 presents the following performance metrics as the training episodes progress: average target received power, agent received power, and tracking error per time-step. As the figure shows, the received power at each target increases with each episode, while the received power at each agent is kept below interference levels after the first half of the episodes. The tracking error, that is the Euclidean distance between the ground truth and the estimated target position, varies between a small range of values and slightly increases over time due to the interference constraint (i.e., in some cases the agents move away from a target in the case another agent already has approached the same target in order to jam it).

A simulated cooperative jamming scenario, where the agents follow their learned policy, is depicted in Fig. 5. The initial agent positions are $[41, 63]^T$ m, $[43, 29]^T$ m and $[41, 27]^T$ m, for agents 1, 2, and 3, respectively, while the initial target states are $[43, -0.30, 26, 0.19, 0.03]^T$ and $[39, 0.39, 60, -0.21, -0.03]^T$ for targets 1 and 2 (in units m, m/s, m, m/s, and rad/s, respectively). The agent and target trajectories and the agent-selected power levels are depicted in Fig. 5(a) and (c), respectively.

The tracking error and the power received at the agents and targets are illustrated in Fig. 5(b) and (d), respectively. As shown from the figures, agent 1 closely follows and jams target 2 with a maximum power level of 0 dBm at each time-step. Moreover, agent 2 starts to follow and jam target 1 with a maximum power level of 0 dBm; this takes place after a few initial time-steps, due to the interference constraints imposed (as initially agents 2 and 3 are closely positioned). Finally, agent 3 remains as close as possible to the targets in order to best estimate their states while avoiding causing (and receiving) interference power; this is the case, since the observed received power at the sa agents is just below the threshold. To evaluate the performance of the proposed approach in this scenario the aforementioned performance metrics are measured by running 1000 episodes under the agents' learned policy and averaging over 50 independent runs. The obtained results indicate a large margin between the target received power (mean: -43.28 dBm, std: -70.43 dBm) and the agent received power (mean: -66.37 dBm, std: -85.96 dBm), with a difference of more than 22 dBm, and a satisfactory tracking error (mean: 13.20 m, std: 0.05 m).

A larger scale scenario considered consists of 6 agents and 6 targets (note that, again, in this larger experiment target states are not estimated). A cooperative jamming simulated scenario, where the agents follow their learned policy, is depicted in Fig. 6. The initial agent positions are $[80, 26]^T$ m, $[11, 43]^T$ m, $[91, 88]^T$ m, $[16, 77]^T$ m, $[62, 84]^T$ m, and $[41, 23]^T$ m, for agents 1, 2, ..., 6, respectively. The initial target states are $[93, -0.40, 86, 0.18, 0.04]^T$, $[44, 0.29, 24, 0.20, 0.03]^T$, $[79, 0.30, 21, 0.19, -0.03]^T$, $[17, -0.29, 80, -0.31, 0.04]^T$, $[62, -0.22, 83, 0.39, 0.04]^T$, and $[13, 0.41, 43, -0.19, 0.03]^T$ for targets 1, 2, ..., 6, respectively, in units m, m/s, m, m/s, and rad/s, respectively. The agents are unable to follow the targets in order to improve jamming and remain relatively motionless, since their received interference power is very close to the critical threshold, as shown in Fig. 6(a) and (c). Only agent 1 is able to properly follow and jam a target (target 3) as it is at a distance from the rest of the agents and receives less interference power.

Therefore, in order to address the interference constraint in the larger scale experiments, a subsequent experiment is performed with agents that now utilize directional antennas for jamming the rogue drones. Outfitting the UAV agents with directional antennas instead of omnidirectional ones can most probably increase the target-received power. This is the case, since the transmitted power is now directed only towards specific areas, allowing for the interference power received at the agents to be reduced and consequently increasing the agent actions that are valid with respect to the interference constraint. A cooperative jamming simulated scenario, where the agents utilize directional antennas and follow their learned policy, is depicted in Fig. 6(e)–(h) (having the same agent and target initializations as in the previous experiment). As observed, the agents are able to successfully follow and jam the targets with maximum power levels, since the agent-received interference power is now significantly decreased.

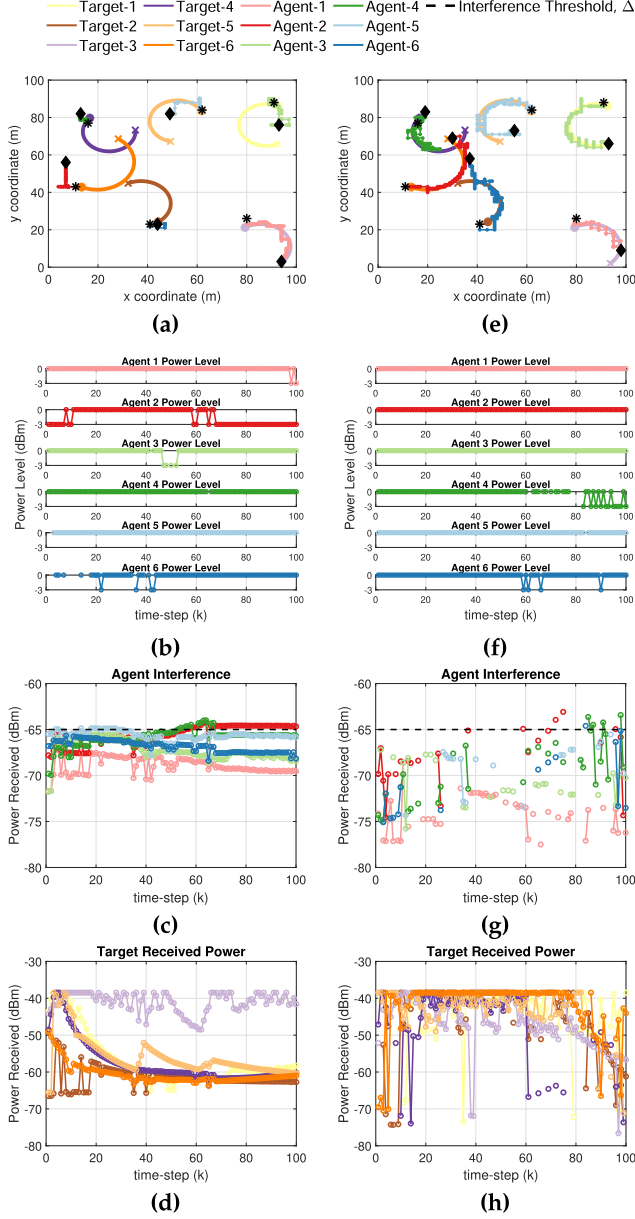


Fig. 6. A simulated scenario using 6 agents and 6 targets for: (i) omnidirectional antennas (a-d) and (ii) directional antennas (e-h).

Additionally, Fig. 7 investigates how the interference constraint in (13) affects the cooperative jamming performance of the system during training, when the agents utilize omnidirectional antennas compared to directional antennas. As expected, the interference constraint significantly limits the actions that the agents with omnidirectional antennas can take, leading to decreased target received power as compared to the case of agents with directional antennas. However, it should be noted that the scenarios utilizing directional antennas increase the agents' action space and consequently reduce the learning speed, thus necessitating the accommodation of more training episodes in order to achieve further improvement.

To demonstrate that the proposed approach scales to larger environments (such as an airport), the simulation environment

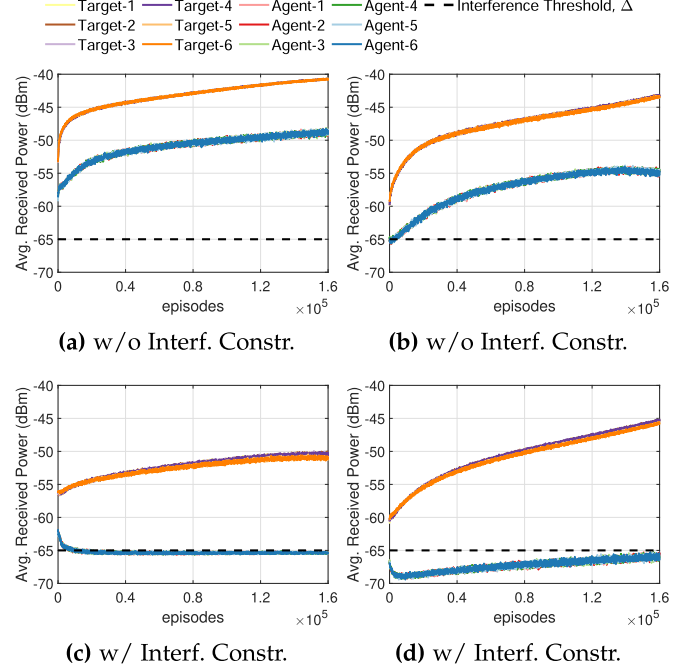


Fig. 7. Impact of the interference constraints, i.e., (13), on the jamming performance for: (i) omnidirectional antennas (a, c) and (ii) directional antennas (b, d).

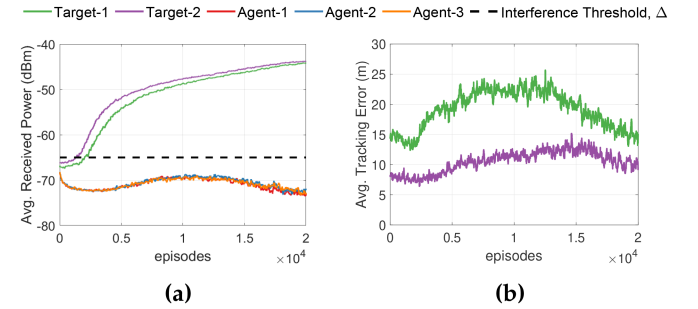


Fig. 8. Training curves for 3 agents and 2 targets in a 1 km $\times$ 1 km area.

is scaled to (1 km $\times$ 1 km) and additional simulations take place for the 3-agent 2-target case, utilizing the same action space, as well as an increased number of time-steps ($K = 400$ time-steps) to allow for larger target trajectories. The results of these experiments are included in Figs. 8 (training curves) and 9 (simulated scenario). For the depicted simulated scenario, the initial agent positions are $[500, 502]^T$ m, $[497, 498]^T$ m, and $[598, 503]^T$ m, for agents 1, 2, and 3, respectively, while the initial target states are $[600, 1.00, 500, 1.00, 0.01]^T$ and $[500, -0.80, 500, -0.90, 0.01]^T$ for targets 1 and 2 (in units of m, m/s, m, m/s, and rad/s, respectively). As shown in Fig. 9, the agents effectively cooperate to jam the targets at each time-step, while keeping the interference power below the critical threshold. More specifically, agent 3 closely follows target 1 and agent 2 closely follows target 2 (the agent and target trajectories overlap in Fig. 9(a), respectively), while agent 1 follows target 1 from a distance to avoid receiving power above interference

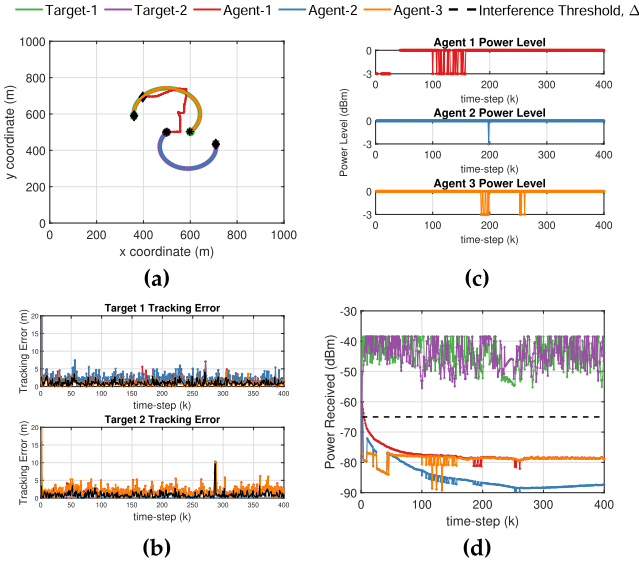


Fig. 9. A simulated scenario using 3 agents and 2 targets in a 1 km $\times$ 1 km area.

levels. The effectiveness of the proposed approach in this scenario is further evaluated by running 1,000 episodes under the agents' learned policy and averaging the following performance metrics over 50 independent runs: the target received power (mean: -43.99 dBm, std: -55.97 dBm), the agent received power (mean: -71.18 dBm, std: -80.23 dBm), and the tracking error (mean: 12.76 m, std: 3.59 m). As demonstrated, the proposed approach effectively scales to larger environments, since the obtained performance metrics are comparable to those of the smaller area scenario (100 m $\times$ 100 m), discussed previously.

C. Evader Target

This section extends the proposed framework to investigate a more challenging target (evader) that reacts to the jamming power it receives from the pursuing agents by altering its mobility dynamics. The evader is set to explore an area and its goal is to reach a certain point of interest (POI), to observe it, and gain as much information about it as possible. At each time-step k , the following reward r_k^j is received by each agent j

$$r_k^j = \begin{cases} wD_{1,k}^j + (1-w)L_{4,k}^j & \text{if (13) is satisfied} \\ -G_{3,k} & \text{otherwise} \end{cases}, \quad (28)$$

where if at least one agent receives interference power above the threshold Δ , the agents are penalized with the global interference reward, $G_{3,k}$; otherwise, the agents' reward comprises of the weighted sum of two sub-rewards, the difference reward for jamming, $D_{1,k}^j$, and the local tracking reward, $L_{4,k}^j = p_D(\tilde{x}_k^j, u_k^j)$, as defined in Sections V-A and IV-C, respectively. As indicated above, the reward function $D_{1,k}^j$ utilizes the predicted target state, $\tilde{x}_k^j$, instead of the ground truth target state, x_k . All rewards are normalized in $[0,1]$ and the parameter w is set to 0.5 .

For the evaluation of the proposed approach the setup described in Section VI-A is utilized, with two agents, one evader

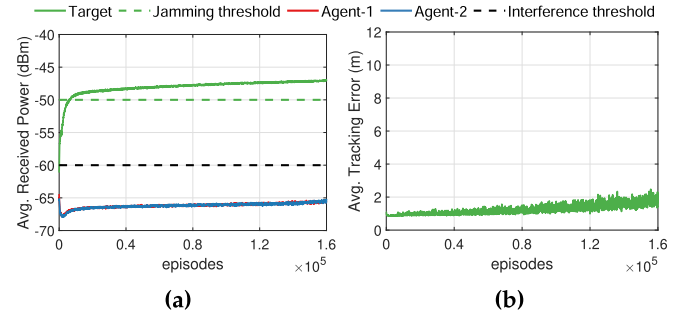


Fig. 10. Training curve for two agents pursuing an evader.

and a POI located at $[75, 75]^T$ m. The evader moves towards the POI with velocity of magnitude equal to $v = \sqrt{\dot{x}^2 + \dot{y}^2} = 1$ m/s, however, if it receives jamming power above a specific threshold (in this case equal to -50 dBm), it moves towards a random destination in order to avoid jamming. In this case, the evader utilizes the noisy dynamical model defined in (3), with x, y initialized between 10 to 90 m, and $\dot{x}, \dot{y}$, and ω taking random initial values in the range of 0.70 to 0.80 m/s and 0.03 to 0.05 rad/s, respectively. The interference threshold Δ is set to -60 dBm and the agent transmitted power levels are selected from $\mathbb{L} = \{\text{off}, -10, 0\}$ dBm. The rest of the parameters are set as in Section VI-A.

As the training curve in Fig. 10 indicates, the agents quickly learn to bring the power received at the evader above the jamming threshold, while keeping their own received power below interference levels. However, as the evader-received power increases, the evader maneuvers with more challenging dynamics, thus the agents' tasks of following the evader and estimating its state become more difficult, leading to an increase in the tracking error. Fig. 11 shows an indicative simulation scenario of the agents' learned policy for the aforementioned scenario. The initial agent positions are $[13, 54]^T$ m, $[8, 49]^T$ m, for agents 1 and 2, respectively, while the initial evader state is $[12, 0.93, 50, 0.37, 0]^T$ in units m, m/s, m, m/s, and rad/s, respectively. The evader maneuvers with steep turns changing directions every few time-steps as shown in Fig. 11(a), where the orange line illustrates the trajectory the evader will take if it does not receive any jamming power (w/o jamming). As demonstrated, agent 1 closely follows the evader and transmits power at maximum levels, while agent 2 keeps the minimum allowable distance from the evader in order to avoid interference. The following performance metrics are obtained under the agents' learned policy, in 1000 episodes and averaged over 50 independent runs: the average number of time-steps the evader is within 5 m from the POI per episode (Fig. 12(a)), the average evader- and agent-received power per time-step (Fig. 12(b)), and the average tracking error per time-step (Fig. 12(c)), with Fig. 12(b) and (c) clearly showing jamming and interference power as well as tracking error, that meet the desired levels. Further, Fig. 12(a) indicates a significant decrease in the number of time-steps the evader is near the POI (approximately one third of the time as compared to the case of no jamming). Finally, it should be noted that for 22.07% of the episodes (in the case of jamming), the evader never approaches

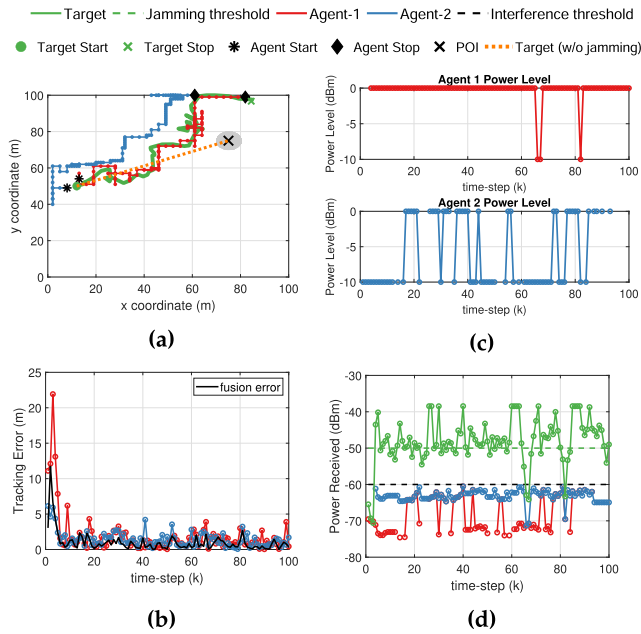


Fig. 11. A simulated scenario using two agents to pursue an evader.

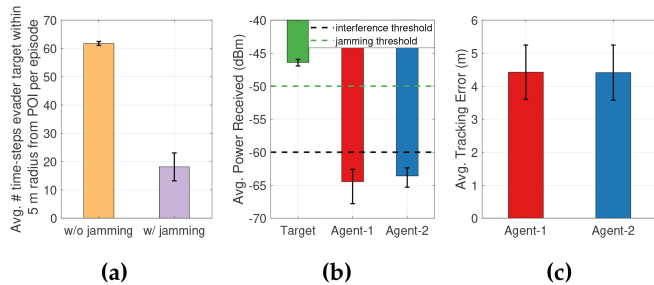


Fig. 12. Performance evaluation results for two agents pursuing an evader: (a) Number of time-steps evader is within 5 m of the POI, (b) Average evader- and agent-received power, and (c) Average tracking error for each agent.

within 5 m of the POI, whereas for the no-jamming scenario this percentage is zero (i.e., the rogue drone always approaches within 5 m of the POI).

VII. CONCLUSION

This work examines the utilization of multiple UAV agents to cooperatively jam multiple rogue drones. An approach based on RL is proposed, utilizing a joint reward function encouraging the coordination between the learning agents for cooperative jamming, that selects the mobility and power control actions for each agent to maximize the jamming power received at the rogue drones, while also keeping the interference power experienced by the agents below a predefined threshold. Performance evaluation results provide an extensive comparison between several reward functions, while demonstrating the superiority of difference rewards in terms of scalability, learning speed, and final joint performance (i.e., in terms of jamming and interference power levels received by the targets and agents, respectively). Nevertheless, the reader should note that scalability is limited

when omnidirectional antennas are placed onboard the agents, due to the presence of the interference constraint. This issue is subsequently addressed by the utilization of directional antennas that mitigate the effect of the interference power experienced by the agents when they are jamming the rogue drones. Finally, an initial attempt is made in this work with targets that perform evasive actions to avoid jamming, with results demonstrating that the proposed technique still allows for the effective jamming and neutralization of the rogue drones.

Future research directions includes the utilization of multi-agent deep RL to address the problem of tracking-and-jamming multiple targets by multiple agents, as well as the consideration of evaders that more intelligently try to evade the agents and avoid being tracked-and-jammed.

REFERENCES

- [1] Federal Aviation Administration, "FAA aerospace forecast: Fiscal years 2022–2042," 2022. [Online]. Available: https://www.faa.gov/sites/faa.gov/files/2022-06/FY2022_42_FAA_Aerospace_Forecast.pdf
- [2] DRONEII, "Drone market report 2021–2026," 2021. [Online]. Available: <https://droneii.com/the-drone-market-in-2021-and-beyond-5-key-takeaways>
- [3] B. Nassi, A. Shabtai, R. Masuoka, and Y. Elovici, "SoK-security and privacy in the age of drones: Threats, challenges, solution mechanisms, and scientific gaps," 2019, *arXiv: 1903.05155*.
- [4] G. Lykou, D. Moustakas, and D. Gritzalis, "Defending airports from UAS: A survey on cyber-attacks and counter-drone sensing technologies," *Sensors*, vol. 20, no. 12, pp. 1–40, 2020.
- [5] K. Wesson and T. Humphreys, "Hacking drones," *Sci. Amer.*, vol. 309, no. 5, pp. 54–59, 2013.
- [6] H. M. Arthur, "Counter-drone systems," Center for the Study of the Drone, Bard College, Tech. Rep., 2018. [Online]. Available: <http://dronecenter.bard.edu/files/2018/02/CSD-Counter-Drone-Systems-Report.pdf>
- [7] P. Valianti, K. Malialis, P. Kolios, and G. Ellinas, "Multi-agent reinforcement learning for multiple rogue drone interception," in *Proc. IEEE Int. Conf. Unmanned Aircr. Syst.*, 2023, pp. 1037–1044.
- [8] S. Park, H. T. Kim, S. Lee, H. Joo, and H. Kim, "Survey on anti-drone systems: Components, designs, and challenges," *IEEE Access*, vol. 9, pp. 42635–42659, 2021.
- [9] H. Kang, J. Joong, J. Kim, J. Kang, and Y. S. Cho, "Protect your sky: A survey of counter unmanned aerial vehicle systems," *IEEE Access*, vol. 8, pp. 168671–168710, 2020.
- [10] X. Shi, C. Yang, W. Xie, C. Liang, Z. Shi, and J. Chen, "Anti-drone system with multiple surveillance technologies: Architecture, implementation, and challenges," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 68–74, Apr. 2018.
- [11] S. R. Ganti and Y. Kim, "Implementation of detection and tracking mechanism for small UAS," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2016, pp. 1254–1260.
- [12] I. Guvenc, F. Koohifar, S. Singh, M. L. Sichitiu, and D. Matolak, "Detection, tracking, and interdiction for amateur drones," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 75–81, Apr. 2018.
- [13] T. Humphreys, "Statement on the security threat posed by unmanned aerial systems and possible countermeasures," The University of Texas at Austin Technical Report, 2015. [Online]. Available: <https://docs.house.gov/meetings/HM/HM09/20150318/103136/HHRG-114-HM09-Wstate-HumphreysT-20150318.pdf>
- [14] T. Multerer et al., "Low-cost jamming system against small drones using a 3D MIMO radar based tracking," in *Proc. Eur. Radar Conf.*, 2017, pp. 299–302.
- [15] A. H. Abunada, A. Y. Osman, A. Khandakar, M. E. H. Chowdhury, T. Khattab, and F. Touati, "Design and implementation of a RF based anti-drone system," in *Proc. IEEE Int. Conf. Informat. IoT Enabling Technol.*, 2020, pp. 35–42.
- [16] N. Souli, I. Theodorou, P. Kolios, and G. Ellinas, "Detection and tracking of rogue UASs using a novel real-time passive radar system," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2022, pp. 576–582.
- [17] N. Souli, P. Kolios, and G. Ellinas, "Onboard passive radar for detection and tracking of rogue UAVs," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2023, pp. 820–826.

- [18] N. Souli et al., "HorizonBlock: Implementation of an autonomous counter-drone system," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2020, pp. 398–404.
- [19] N. Souli, P. Kolios, and G. Ellinas, "An autonomous counter-drone system with jamming and relative positioning capabilities," in *Proc. IEEE Int. Conf. Commun.*, 2022, pp. 5110–5115.
- [20] A. Barisic et al., "Multi-robot system for autonomous cooperative counter-UAS missions: Design, integration, and field testing," in *Proc. IEEE Int. Symp. Safety Security Rescue Robot.*, 2022, pp. 203–210.
- [21] V. U. Castrillo, A. Manco, D. Pascarella, and G. Gigante, "A review of counter-UAS technologies for cooperative defensive teams of drones," *Drones*, vol. 6, no. 3, pp. 1–36, 2022.
- [22] A. Guerra, D. Dardari, and P. M. Djuric, "Dynamic radar networks of UAVs: A tutorial overview and tracking performance comparison with terrestrial radar networks," *IEEE Veh. Technol. Mag.*, vol. 15, no. 2, pp. 113–120, Jun. 2020.
- [23] J. Rothe, M. Strohmeier, and S. Montenegro, "A concept for catching drones with a net carried by cooperative UAVs," in *Proc. IEEE Int. Symp. Safety Security Rescue Robot.*, 2019, pp. 126–132.
- [24] M. R. Brust, G. Danoy, P. Bouvry, D. Gashi, H. Pathak, and M. P. Gonçalves, "Defending against intrusion of malicious UAVs with networked UAV defense swarms," in *Proc. IEEE 42nd Conf. Local Comput. Netw. Workshops*, 2017, pp. 103–111.
- [25] P. Valianti, S. Papaioannou, P. Kolios, and G. Ellinas, "Multi-agent coordinated close-in jamming for disabling a rogue drone," *IEEE Trans. Mobile Comput.*, vol. 21, no. 10, pp. 3700–3717, Oct. 2022.
- [26] P. Valianti, S. Papaioannou, P. Kolios, and G. Ellinas, "Multi-agent coordinated interception of multiple rogue drones," in *Proc. IEEE Glob. Commun. Conf.*, 2020, pp. 1–6.
- [27] S. Papaioannou, P. Kolios, and G. Ellinas, "Distributed estimation and control for jamming an aerial target with multiple agents," *IEEE Trans. Mobile Comput.*, vol. 22, no. 12, pp. 7203–7217, Dec. 2023.
- [28] N. Souli, P. Kolios, and G. Ellinas, "Multi-agent system for rogue drone interception," *IEEE Robot. Automat. Lett.*, vol. 8, no. 4, pp. 2221–2228, Apr. 2023.
- [29] Y. Song, M. Steinweg, E. Kaufmann, and D. Scaramuzza, "Autonomous drone racing with deep reinforcement learning," in *Proc. IEEE Int. Conf. Intell. Robots Syst.*, 2021, pp. 1205–1212.
- [30] X. Wang, M. Cen Gursoy, T. Erpek, and Y. E. Sagduyu, "Jamming-resilient path planning for multiple UAVs via deep reinforcement learning," in *Proc. IEEE Int. Conf. Commun. Workshops*, 2021, pp. 1–6.
- [31] M. Theile, H. Bayerlein, R. Nai, D. Gesbert, and M. Caccamo, "UAV coverage path planning under varying power constraints using deep reinforcement learning," in *Proc. IEEE Int. Conf. Intell. Robots Syst.*, 2020, pp. 1444–1449.
- [32] A. Palamas, N. Souli, T. Panayiotou, P. Kolios, and G. Ellinas, "A multi-task learning framework for drone state identification and trajectory prediction," in *Proc. IEEE Int. Conf. Distrib. Comput. Smart Syst. Internet Things 5th Int. Workshop Wireless Sensors Drones Internet Things*, 2023, pp. 1–8.
- [33] L. S. B. Vlahov, E. Squires, and C. Pippin, "On developing a UAV pursuit-evasion policy using reinforcement learning," in *Proc. IEEE Int. Conf. Mach. Learn. Appl.*, 2019, pp. 859–864.
- [34] Y. Wang, L. Dong, and C. Sun, "Cooperative control for multi-player pursuit-evasion games with reinforcement learning," *Neurocomputing*, vol. 412, pp. 101–114, 2020.
- [35] W. Du, T. Guo, J. Chen, B. Li, G. Zhu, and X. Cao, "Cooperative pursuit of unauthorized UAVs in urban airspace via multi-agent reinforcement learning," *Transp. Res. Part C: Emerg. Technol.*, vol. 128, 2021, Art. no. 103122.
- [36] J. Moon, S. Papaioannou, C. Laoudias, P. Kolios, and S. Kim, "Deep reinforcement learning multi-UAV trajectory control for target tracking," *IEEE Internet of Things J.*, vol. 8, no. 20, pp. 15441–15455, Oct. 2021.
- [37] M. Min, L. Xiao, D. Xu, L. Huang, and M. Peng, "Learning-based defense against malicious unmanned aerial vehicles," in *Proc. IEEE 87th Veh. Technol. Conf.*, 2018, pp. 1–5.
- [38] R. S. Sutton and A. G. Barto, *Introduction to Reinforcement Learning*. Cambridge, MA, USA: MIT Press, 1998.
- [39] M. Wooldridge, *An Introduction to Multiagent Systems*. Hoboken, NJ, USA: Wiley, 2009.
- [40] L. Busoni, R. Babuska, and B. De Schutter, "A comprehensive survey of multiagent reinforcement learning," *IEEE Trans. Syst. Man Cybern., Part C (Appl. Rev.)*, vol. 38, no. 2, pp. 156–172, Mar. 2008.
- [41] C. Claus and C. Boutilier, "The dynamics of reinforcement learning in cooperative multiagent systems," in *Proc. Conf. Artif. Intell./Innov. Appl. Artif. Intell.*, 1998, pp. 746–752.
- [42] K. Malialis and D. Kudenko, "Distributed response to network intrusions using multiagent reinforcement learning," *Eng. Appl. Artif. Intell.*, vol. 41, pp. 270–284, 2015.
- [43] D. H. Wolpert, K. R. Wheeler, and K. Tumer, "Collective intelligence for control of distributed dynamical systems," *Europhysics Lett.*, vol. 49, no. 6, pp. 708–714, 2000.
- [44] L. Yliniemi, A. K. Agogino, and K. Tumer, "Multirobot coordination for space exploration," *AI Mag.*, vol. 35, no. 4, pp. 61–74, 2014.
- [45] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation With Applications to Tracking and Navigation: Theory Algorithms and Software*. Hoboken, NJ, USA: Wiley, 2004.
- [46] M. Roth, G. Hendeby, and F. Gustafsson, "EKF/UKF maneuvering target tracking using coordinated turn models with polar/cartesian velocity," in *Proc. Int. Conf. Inform. Fusion*, 2014, pp. 1–8.
- [47] B. Mora, A. V. Moll, I. Weintraub, D. Casbeer, and A. Chakravarthy, "Escape from an orbiting pursuer with a nonzero capture radius," 2023, *arXiv:2310.01203*.
- [48] A. Tsiamis, A. Karapetyan, Y. Li, E. C. Balta, and J. Lygeros, "Predictive linear online tracking for unknown targets," 2024, *arXiv:2402.10036*.
- [49] Remote identification will become mandatory for drones across Europe, EASA, 2023. [Online]. Available: <https://www.easa.europa.eu/en/document-library/general-publications/remote-identification-will-become-mandatory-drones-across#:text=The drone operator is required>
- [50] P.-H. Chen et al., "Integration time and time-bin width in ranging histograms for a photon-counting LiDAR," *IEEE/OPTICA J. Lightw. Technol.*, vol. 40, no. 17, pp. 5870–5875, Sep. 2022.
- [51] Z. Chen, "Bayesian filtering: From Kalman filters to particle filters, and beyond," Adaptive Syst. Lab., McMaster University, Tech. Rep., 2003. [Online]. Available: <https://www.scribd.com/doc/187752564/Bayesian-Filtering-From-Kalman-Filters-to-Particle-Filters-and-Beyond>
- [52] R. P. S. Mahler, *Advances in Statistical Multisource-Multitarget Information Fusion*. Norwood, MA, USA: Artech House, 2014.
- [53] P. O. Arambel, C. Rago, and R. K. Mehra, "Covariance intersection algorithm for distributed spacecraft state estimation," in *Proc. IEEE Amer. Control Conf.*, 2001, pp. 4398–4403.
- [54] M. S. Arulampalam, S. Maskell, N. Gordon, and T. Clapp, "A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking," *IEEE Trans. Signal Process.*, vol. 50, no. 2, pp. 174–188, Feb. 2002.



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